NTIRE 2026: Robust AI-Generated Image Detection in the Wild

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Abstract

The rapid evolution of text-to-image models like FLUX.1 and SDXL has made synthetic imagery indistinguishable from real photography, particularly under "in-the-wild" transformations like compression and blurring. Participating in the NTIRE 2026 challenge, we propose a hybrid detection framework that fuses spatial RGB features with a frequency-domain analysis branch. Our system employs a modified EfficientNet-B5 backbone using a four-channel input tensor ($3_{\text{RGB}} + 1_{\text{FFT}}$), where the fourth channel consists of a log-magnitude Fast Fourier Transform (FFT) spectrum calculated on-the-fly on the GPU. To maximize generalization and the primary Robust ROC AUC metric, we utilize a stochastic augmentation pipeline featuring twelve transformations with Gaussian-sampled severity. This dual-domain solution remains computationally efficient, strictly adhering to the mandated 1 FPS inference speed and 80GB GPU memory constraints.

Keywords: AI-Generated Images, Image Forensics, Deep Learning, NTIRE 2026

1. Introduction

The rapid proliferation of high-fidelity text-to-image (T2I) generative models, such as FLUX.1, SDXL, and Janus Pro, has made synthetic imagery nearly indistinguishable from authentic photography in many real-world scenarios. While these models offer unprecedented creative potential, they simultaneously pose severe risks to digital trust, forensic integrity, and content safety. A critical challenge in modern AI detection is the "in-the-wild" environment, where images are routinely subjected to transformations such as JPEG compression, resizing, blurring, and cropping. These common post-processing operations often strip away the subtle pixel-level signatures that earlier detection methods relied upon, making robust identification a significant hurdle for current forensic systems.

Historically, AI detection approaches primarily focused on capturing spatial inconsistencies or specific architectural artifacts left behind by early Generative Adversarial Networks (GANs). These methods often utilized standard Convolutional Neural Networks (CNNs) to identify local textures or frequency-agnostic features unique to a specific generator. However, these spatial-only techniques proved to be fragile; they frequently suffered from poor generalization to unseen generative architectures and were easily bypassed by simple image degradations. Because these previous approaches were often over-optimized for specific generator "fingerprints," they failed to maintain reliability when faced with the diverse distribution shifts encountered on social media platforms and digital distribution channels.

The NTIRE 2026 Challenge exemplifies the shift toward ensuring high generalization and robustness by prioritizing "Robust ROC AUC" as the primary evaluation metric. Unlike previous benchmarks, this challenge requires detectors to remain accurate across a massive training dataset of $\sim$ 277K images generated by 20 different models and subjected to a wide array of "in-the-wild" style transformations. Furthermore, the challenge imposes strict operational constraints: submitted models must fit within an 80 GB GPU memory limit and maintain an inference speed of no slower than 1 Frame Per Second (FPS). Consequently, the field is moving toward hybrid architectures that can efficiently identify permanent forensic markers while adhering to rigorous computational limits.

To address these contemporary challenges, we propose a hybrid detection framework that fuses spatial and frequency-domain analysis using an EfficientNet-B5 backbone. Our approach moves beyond traditional RGB-only analysis by introducing a 4-channel input tensor ($3_{\text{RGB}} + 1_{\text{FFT}}$) that incorporates a log-magnitude spectrum generated via a GPU-accelerated 2D Fast Fourier Transform (FFT) branch. By calculating the FFT on the GPU during the forward pass, our model can efficiently identify periodic synthetic artifacts—such as grid-like patterns or spectral peaks—that persist even after heavy spatial blurring or compression. This dual-domain representation, combined with a stochastic distortion training pipeline utilizing a 60% probability per image, allows our system to maximize Robust ROC AUC scores while strictly complying with the efficiency requirements of the NTIRE 2026 challenge.

2. Proposed Methodology

Our proposed framework adopts a dual-domain approach, leveraging both spatial and frequency-domain features to enhance detection robustness. The overall architecture, as illustrated in Fig. 1, integrates an EfficientNet-B5 backbone with a GPU-accelerated frequency branch to capture generative artifacts that persist under heavy image degradation.

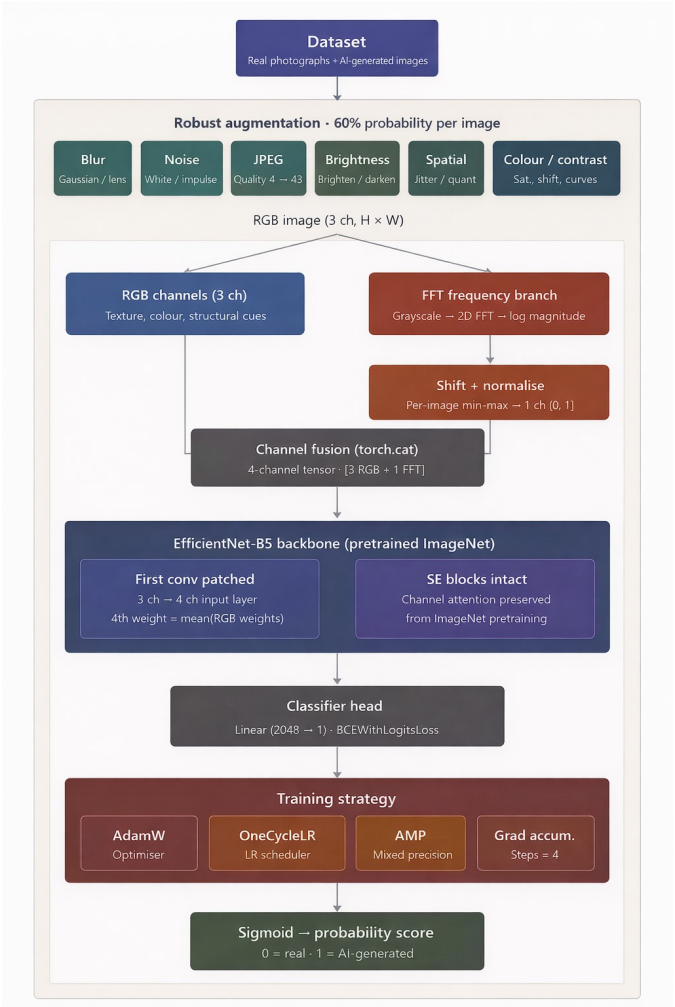


Figure 1: Architecture diagram of the proposed Robust AI-Generated Image Detection framework. The system efficiently fuses RGB and FFT features before processing through the modified EfficientNet-B5 stem.

2.1. Robust Augmentation Strategy

To optimize for the primary Robust ROC AUC metric, we implement a stochastic distortion pipeline that applies transformations with a 60% probability per image during training. The pipeline selects 1 to 3 random distortions from seven distinct groups: blur (Gaussian/Lens), noise (White/Impulse), JPEG compression (quality 4–43), color distortions, brightness changes, spatial jitter, and tonal quantization. Notably, the severity of these distortions is sampled from a Gaussian distribution centered at mild levels to ensure generalization to realistic "in-the-wild" scenarios.

2.2. Frequency Domain Branch

A core component of our methodology is the on-the-fly computation of frequency-domain features within the model's forward pass, ensuring compliance with the 1 FPS inference constraint. For each input RGB image, we compute a grayscale map G using the luminosity method. This is followed by a 2D Fast Fourier Transform (FFT) and a frequency shift to center the DC component. We then extract the log-magnitude spectrum to compress the dynamic range:

$$S = \log\left(|\mathrm{FFT}_{\mathrm{shift}}(G)| + 10^{-6}\right) \quad (1)$$

The resulting spectrum is min-max normalized per-image to a $[0, 1]$ range, creating a 1-channel feature map that highlights periodic upsampling artifacts common in generative models.

2.3. Architectural Modifications and Fusion

The spatial RGB branch (3 channels) and frequency branch (1 channel) are concatenated into a 4-channel tensor of shape $(B, 4, 456, 456)$. To process this input, we replace the initial convolutional layer of a pretrained EfficientNet-B5 backbone with a 4-channel variant. We initialize the first three channels using pretrained ImageNet weights and set the 4th (FFT) channel as the mean of the RGB weights. This preserves the Squeeze-and-Excitation (SE) blocks' ability to perform channel-wise attention, dynamically weighing spatial versus frequency cues.

2.4. Training and Optimization

Training is performed using *BCEWithLogitsLoss* and the AdamW optimizer with a learning rate of 10^{-4}. We utilize a *OneCycleLR* scheduler to facilitate faster convergence and escape local minima during the early stages of training. To accommodate the 16 GB VRAM limit of the Tesla P100 hardware, we employ Automatic Mixed Precision (AMP) and a gradient accumulation strategy with 8 steps, achieving an effective batch size of 32.

3. Dataset and Evaluation Metrics

3.1. Dataset Description

The NTIRE 2026 dataset is specifically curated to evaluate the robustness of AI-generated image detectors against "in-the-wild" distortions. The data is partitioned into five primary splits to facilitate training and benchmarking, as summarized in Table 1.

Table 1: Dataset Split Statistics and Complexity

Split	# Images	Ratio (R:F)	Models	Transf.
Train	~277K	~1:1.77	20	12
Validation	10K	1:1	9	19
Validation Hard	2.5K	1:1	7	19
Test (Public)	2.5K	1:1	10	22
Test (Private)	2.5K	~1:1	10	24

The dataset features a transition from 12 basic distortions in training to 24 complex perturbations in the test phase, including Neural Image Compression (JPEG AI) and various Watermark Attacks.

3.2. Evaluation Metrics

The primary objective is to measure the detector's global discriminative ability and its robustness to post-processing. The performance is evaluated using the following metrics:

- **Primary Metric: Robust ROC AUC**
  Calculated across all images *after* transformations. This measures the model's ability to maintain performance despite heavy image degradation.
- **Secondary Metric: Clean ROC AUC**
  Calculated across all images *without* transformations to assess the base discriminative power of the detector.

3.2.1. Metric Formalization

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. These are defined as:

$$TPR = \frac{TP}{TP + FN} \quad , \quad FPR = \frac{FP}{FP + TN} \tag{2}$$

The Area Under the Curve (AUC) represents the probability that a randomly chosen generated image receives a higher score than a randomly chosen real image. It is formally calculated as:

$$AUC = \int_0^1 TPR(FPR)\, d(FPR) \tag{3}$$

An AUC of 1.0 indicates perfect separation, while 0.5 represents random guessing. This metric is threshold-independent and robust to the class imbalance observed in the training split.

3.3. System Constraints

For the final evaluation phase, the proposed models must adhere to strict computational constraints:

1. **Memory:** Maximum GPU memory consumption of 80 GB.
2. **Latency:** Inference speed must be no slower than 1 frame per second (FPS).

4. Experimentation and Results

4.1. Experimental Setup

The major hyperparamter configurations are detailed in Table 2.

Table 2: Training Hyperparameters and Model Configuration

Parameter	Value
Backbone	EfficientNet-B5 (4-Channel)
Input Resolution	456×456
Batch Size	4 (Accumulation Steps: 8)
Effective Batch Size	32
Epochs	4
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-2}
Loss Function	BCE with Logits
Scheduler	OneCycleLR
Distortion Prob.	0.4
Hardware	Tesla P100 (16GB VRAM)

5. Results

5.1. Performance Comparison of Model Variants

To identify the most effective architecture for the NTIRE challenge, we conducted an ablation study comparing various EfficientNet backbones and feature fusion strategies. The results, summarized in Table 3, evaluate each variant's ability to maintain high discriminative performance across clean and distorted datasets.

Table 3: Ablation Study and Performance Comparison of Model Architectures.

Model Variant	Clean AUC	Robust AUC
EfficientNet-B0	0.9290	0.8402
EfficientNet-B4	0.9410	0.9212
Swin Transformer	0.9709	0.8965
EffNet-B5 (RGB + FFT)	**0.9768**	**0.9341**
EffNet-B5 (PID, FFT, Cont. Loss)	0.9506	0.8701

As shown in the comparison, the 4-channel EfficientNet-B5 (RGB + FFT) architecture achieved the highest performance in both categories. While the Swin Transformer showed strong baseline (Clean) performance, it lacked the robustness of the B5-FFT fusion when subjected to the distortion pipeline. Interestingly, the addition of contrastive loss and PID features did not yield superior results in this specific task, likely due to the increased complexity of the optimization landscape. Consequently, the EfficientNet-B5 with direct FFT-log magnitude concatenation was selected as our final submission.

5.2. Validation Phase

During the validation phase, our model was evaluated among 193 participants. Our best-performing submission, utilizing the EfficientNet-B5 architecture, achieved the 100th position on the leaderboard. The model yielded a Clean ROC AUC of 0.9220 and a Robust ROC AUC of 0.8324, as shown in Fig. 2. Experimental comparisons during this phase indicated that the EfficientNet-B5 backbone consistently outperformed alternative architectures, including EfficientNet-B4 and Tiny-ViT, particularly in maintaining higher robust scores under the standard 12-stage distortion pipeline.

5.3. Testing Phase

In the final testing phase, which consisted of 94 active participants, our framework secured the 75th position. As illustrated in Fig. 3, the model achieved a Clean ROC AUC of 0.8084 and a Robust ROC AUC of 0.7099. We observed a relative decrease in performance compared to the validation phase; this is primarily attributed to the increased complexity of the testing dataset, which introduced up to 24 distinct transformations and

96	ImSecuGroup3DipGosGauKim	2026-02-27 19:32	569816	0.9343	0.9149	0.8717	**0.83**
97	lingxiang	2026-03-05 22:23	594666	0.9933	0.9644	0.9818	**0.83**
98	seventeen_17	2026-03-07 07:52	598988	0.9676	0.9373	0.9087	**0.83**
99	mgiant	2026-02-19 14:45	549242	0.9942	0.9597	0.9781	**0.83**
100	sharan	2026-02-20 15:49	552476	0.9768	0.9341	0.922	**0.83**
101	cuijy	2026-03-07 07:33	598938	0.9883	0.9201	0.9758	**0.82**
102	tianye	2026-03-04 10:31	589139	0.9944	0.9205	0.9892	**0.80**
103	rlghksdbs	2026-03-04 14:54	589854	0.9867	0.9267	0.9571	**0.80**
104	d4darius	2026-02-19 06:45	548760	0.9711	0.9344	0.8938	**0.79**

Figure 2: Leaderboard standing during the Validation Phase. Our team (sharan) secured the 100th position out of 193 participants.

proprietary generative models not present in the initial training or validation splits. Despite the more aggressive distortion environment, the dual-domain (RGB+FFT) approach ensured the model remained functional and surpassed the random-guess baseline (0.5).

73	group1bgpctdnw	2026-03-15 18:41	626034	n/a	n/a	0.7777	**0.7159**
74	ImSecu_group7_alexaye	2026-03-16 02:20	627088	n/a	n/a	0.8319	**0.7108**
75	sharan	2026-03-15 19:43	626238	n/a	n/a	0.8084	**0.7099**
76	d4darius	2026-03-10 19:26	610994	n/a	n/a	0.8145	**0.7012**
77	aaitdads	2026-03-11 01:34	611822	n/a	n/a	0.9228	**0.6994**

Figure 3: Final standing in the Testing Phase leaderboard, showing a Robust ROC AUC of 0.7099.

6. Conclusion

7. Conclusion and Future Work

In this project, we developed a robust AI-generated image detection framework for the NTIRE 2026 challenge, demonstrating that a dual-domain approach—integrating spatial RGB features with GPU-accelerated FFT magnitude spectra—provides a resilient defense against synthetic forgeries even under significant post-processing degradation. While our EfficientNet-B5 architecture achieved the peak validation accuracy , the transition to a more complex testing environment with 24 distinct transformations highlighted the persistent challenge of model generalization in “in-the-wild” scenarios. Future work will focus on closing the performance gap between validation and unseen testing data by transitioning to more advanced, lightweight backbones like EfficientNet-V2 to improve inference latency, expanding the stochastic augmentation pipeline to include a wider variety of proprietary generative artifacts, and exploring deeper attention-based fusion techniques to dynamically weigh spatial and frequency cues. By refining these strategies, we aim to develop a more universal detector capable of maintaining high Robust ROC AUC scores across diverse and evolving generative models.